Estimating methane from rice under different water and straw management: an IPCC Tier 1 model with uncertainty

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Summary

Rice fields that stay flooded for most of the season are a major source of methane. This project built a Python model of methane from one rice crop using the Tier 1 method of the IPCC 2019 Refinement and applied it to South Asia under four scenarios: continuous flooding and alternate wetting and drying (AWD), each with and without 5 t ha^{-1} of straw mixed in shortly before planting. To test how uncertain the answer is, the calculation was repeated 10,000 times with values picked at random from the IPCC ranges (a Monte Carlo analysis). Flooding gave a middle value of 95.1 kg CH_4 ha^{-1} per season, with 95% of trials between 56.0 and 163.5. AWD gave 52.5 (30.7 to 89.9), a saving of 45% (18% to 63%). Fresh straw multiplied emissions by 2.88. The wide ranges show why field measurements are needed to narrow estimates.

1. 1. Introduction

Rice is grown in flooded fields, and the standing water keeps the soil short of oxygen. Microbes then decompose organic matter and release methane, a strong greenhouse gas (Banja et al., 2026). Banja et al. (2026) put rice at about 8% of human-caused methane emissions. Wang et al. (2023), using the 2019 IPCC method, estimated global emissions from rice fields at 27 ± 6 Tg CH_4 per year for 2008 to 2017, and concluded that better water and straw management could remove about one third of them.

Two practices matter most here. In alternate wetting and drying (AWD) the field is flooded and then allowed to dry before it is flooded again. Straw, if it is mixed into the soil shortly before planting, adds easily decomposed carbon and raises emissions (IPCC, 2019). Studies that combined the results of many field experiments report large methane reductions when fields are not kept flooded all the time: 53% in Jiang et al. (2019) and 51.6% for AWD in Zhao et al. (2024).

Many regions have no field measurements for every combination of practices, so inventories use the IPCC Tier 1 method. It multiplies a baseline daily emission factor by scaling factors for water regime, pre-season water and organic amendments (IPCC, 2019). The method is simple enough to program, which makes it a good way to see how each practice changes the estimate and how uncertain the defaults are.

This project built that calculation in Python and asked one question: how much does methane from one rice crop in South Asia change under AWD, with and without straw, according to the IPCC defaults, and how uncertain is that estimate?

2. Methods

2.1 Model

The model follows Section 5.5 of Volume 4, Chapter 5 of the IPCC 2019 Refinement (IPCC, 2019). The adjusted daily emission factor is

$$EFi = EFc \times SFw \times SFp \times SFo \quad \text{(IPCC Equation 5.2)}$$

where EFc is the baseline emission factor for continuously flooded fields without organic amendments (kg CH_4 ha^{-1} d^{-1}), SFw and SFp are scaling factors for the water regime during and before the cultivation period, and SFo is the scaling factor for organic amendments. The amendment factor is

$$SFo = (1 + \Sigma\ ROAi \times CFOAi) \wedge 0.59 \quad \text{(IPCC Equation 5.3)}$$

where ROAi is the application rate of amendment i (t ha^{-1}) and CFOAi is its conversion factor. Seasonal methane per hectare is EFi multiplied by the cultivation period in days. Equation 5.1 scales this to national totals and was not needed here.

2.2 Scenarios and parameters

Four scenarios were run for South Asia: continuous flooding, AWD, and each of these with straw at 5 t ha^{-1} mixed into the soil less than 30 days before planting. The IPCC places AWD in the class “multiple drainage periods” (Table 5.12). The field was taken to be without standing water for less than 180 days before the crop, which is the condition the baseline factor assumes (Table 5.11). Table 1 lists every value and its source.

Parameter	Value (IPCC range)	IPCC 2019 source
Baseline emission factor EFc, South Asia	0.85 kg CH_4 ha^{-1} d^{-1} (0.58 to 1.26)	Table 5.11, p. 5.53
Cultivation period, South Asia	112 d (90 to 140)	Table 5.11A, p. 5.53
SFw, continuously flooded	1.00 (0.73 to 1.27)	Table 5.12, p. 5.54
SFw, multiple drainage periods (AWD)	0.55 (0.41 to 0.72)	Table 5.12, p. 5.54
SFp, non-flooded pre-season < 180 d	1.00 (0.88 to 1.12)	Table 5.13, p. 5.55
CFOA, straw incorporated < 30 d before planting	1.00 (0.85 to 1.17)	Table 5.14, p. 5.56
Exponent in SFo	0.59 (0.54 to 0.64)	Equation 5.3, p. 5.55

Table 1. Parameters used in the model.

2.3 Uncertainty analysis

To see how uncertain the estimate is, the calculation was repeated 10,000 times (a Monte Carlo analysis). In each trial, every factor was picked at random from its IPCC range, with values close to the IPCC default more likely than values at the ends of the range. In each trial the baseline factor, the season length and the pre-season factor were the same for the flooded and AWD scenarios, so only the water factor differed between them. The code was written in Python with NumPy, pandas and matplotlib. As a check, it gives the same answer as the IPCC worked example in Section 5.5.4 (410.47 Gg CH_4 per year).

3. Results

3.1 Central estimates

With the default factors, continuous flooding gave 95.2 kg CH_4 ha^{-1} per season (0.85 kg ha^{-1} d^{-1} over 112 days). AWD gave 52.4, a reduction of 45%. Straw at 5 t ha^{-1} raised the flooded estimate to 274.0 (+188%) because SFo was 2.88. AWD with the same straw gave 150.7, which is 58% above flooding without straw (Table 2, Figure 1).

Scenario	CH_4 (kg ha^{-1} per season)	Change vs flooded
Flooded	95.2	0%
AWD	52.4	−45%
Flooded + straw 5 t ha^{-1} (< 30 d)	274.0	+188%
AWD + straw 5 t ha^{-1} (< 30 d)	150.7	+58%

Table 2. Estimated methane per hectare for one rice crop, South Asia, IPCC 2019 Tier 1 defaults.

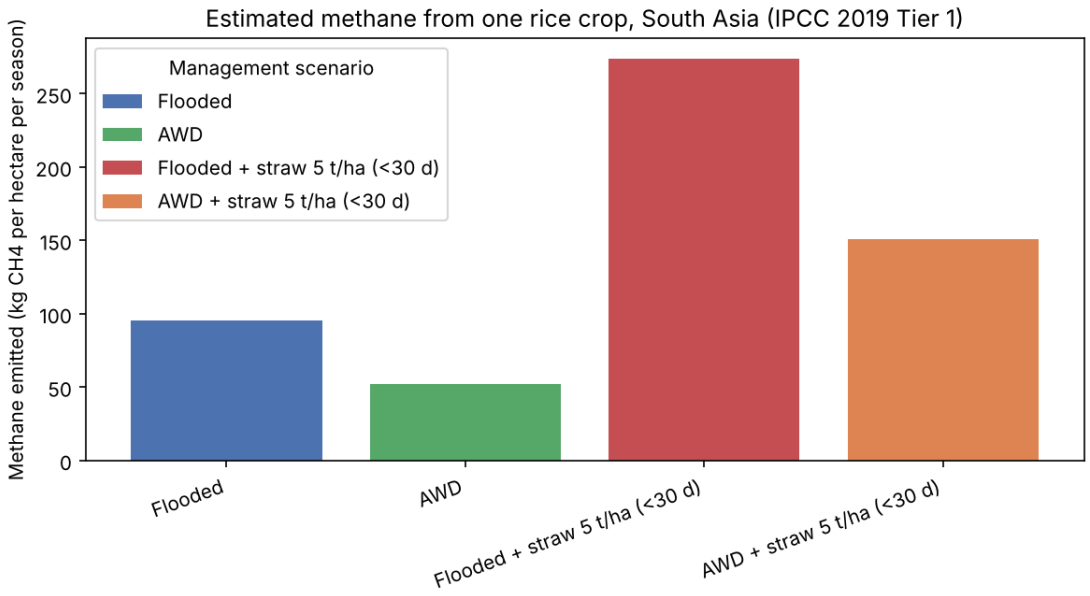


Figure 1. Estimated methane from one rice crop in South Asia under four management scenarios. Each colour is one scenario, named in the legend. Taller bars mean more methane.

3.2 Uncertainty

Across the 10,000 trials, the middle value (median) for flooded rice was 95.1 kg CH_4 ha^{-1}, and 95% of trials fell between 56.0 and 163.5. For AWD the median was 52.5, with 95% of trials between 30.7 and 89.9. The two ranges overlap. However, both scenarios used the same baseline values within each trial, so AWD came out lower in nearly every trial: 95% of trials gave a saving between 18% and 63%, with a median of 45% (Table 3, Figure 2). Straw scenarios were not included in this run.

Quantity	Median	Range covering 95% of trials
Flooded (kg CH_4 ha^{-1} per season)	95.1	56.0 to 163.5
AWD (kg CH_4 ha^{-1} per season)	52.5	30.7 to 89.9
AWD saving relative to flooded (%)	45	18 to 63

Table 3. Monte Carlo results (10,000 trials).

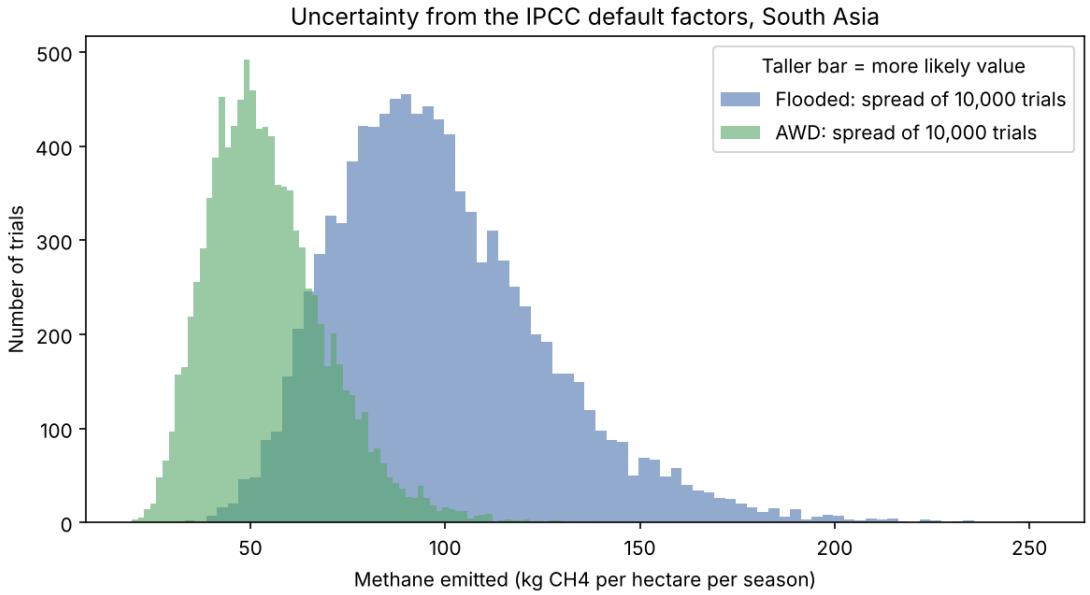


Figure 2. Spread of 10,000 trials for flooded (blue) and AWD (green) rice. A taller bar means that value came up more often.

4. Discussion

The central estimate is a 45% methane saving under AWD, with a 95% range of 18% to 63%. This value comes entirely from the IPCC scaling factor for multiple drainage periods (0.55), because the model uses no measurements from a real field.

Measured results sit close to this. Jiang et al. (2019) found a 53% methane reduction under non-continuous flooding, Zhao et al. (2024) found 51.6% for AWD, and Balaine et al. (2019) reported 41% to 71% depending on how dry the soil became. The default saving is therefore a little below the average of the measurements but inside their range. Jiang et al. (2019) also suggested a factor of 0.36 for multiple drying events, lower than the 0.55 used here. A factor of 0.36 with the same baseline would imply a saving of about 64%.

Straw changed the result more than water management did. The IPCC notes that emissions rise with the amount of amendment applied and that timing matters (IPCC, 2019). Straw mixed in less than 30 days before planting has a conversion factor of 1.00, which gave an SFo of 2.88 for 5 t ha^{-1}. With the factor for straw applied more than 30 days earlier (0.19), SFo would be 1.48 and the AWD scenario would give 77.6 kg CH_4 ha^{-1} instead of 150.7. In this model, when straw goes into the soil is as important as how the water is managed.

The pattern fits the mechanism. Methane comes from decomposition in flooded, oxygen-poor soil (Banja et al., 2026). Periods without standing water cut the time the soil stays in that state, and fresh straw supplies easily decomposed carbon that feeds it.

A methane-only model leaves out trade-offs. Zhao et al. (2024) found that AWD cut methane by 51.6% but raised nitrous oxide by 44.0%, and the combined warming effect still fell by 46.9%. Jiang et al. (2019) found yield fell by 3.6% on average when fields were not kept flooded all the time. I would like to narrow the range by measuring methane and nitrous oxide directly in a field with closed chambers, sampled every week through the season and again after each drainage and re-flooding. I would also record the water level and soil redox potential (a measure of how short of oxygen the soil is), the date straw is added and the grain yield. These readings would replace the IPCC defaults for the water and straw factors with local values, and they would show whether the methane saving under AWD comes at the cost of extra nitrous oxide or lower yield.

5. Limitations

- The factors are IPCC defaults, not measurements, so results describe typical conditions and not any particular field.
- Only the Tier 1 method was used. Soil type and rice variety factors have no IPCC defaults, so they are not in the model.
- Each factor was picked at random on its own, apart from the ones shared between scenarios. In real fields some factors may rise and fall together.
- The IPCC tables describe the ranges as 95% confidence intervals, but Section 5.5.5 calls them standard deviations. I treated them as 95% intervals. A different reading would change the spread.
- The straw scenarios were not included in the Monte Carlo run.
- The model covers methane only. It does not include nitrous oxide, yield or water use.

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to 5.14 of the IPCC (2019) Refinement, and rewrote the text in my own words. The choice of scenarios, the interpretation of the results and any remaining errors are my own.

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